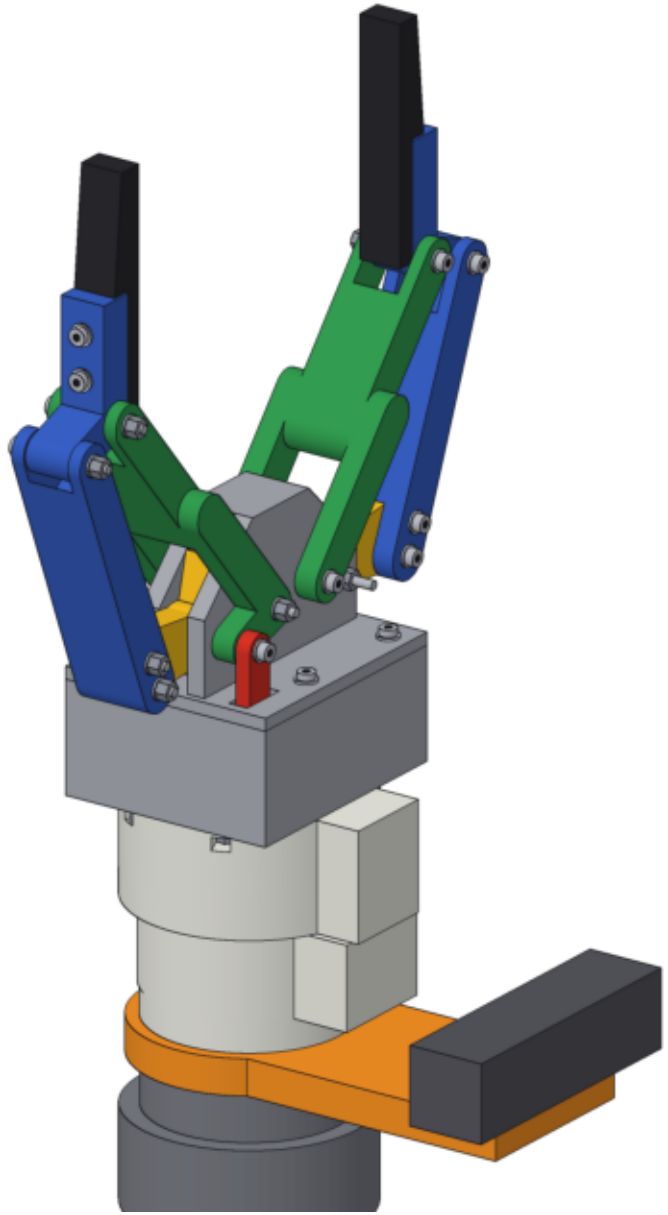
HomeMade2F140

Assembly manual



2-finger gripper with the Robotiq 2F-140 kinematics

DFRobot SER0062 (D5545GS) servo - version v2

17 printed parts - 36 M3 screws - 4 M6 bolts

30 M3 nyloc nuts - 50 M3 washers

Tools: 2.5 mm and 5 mm hex keys,

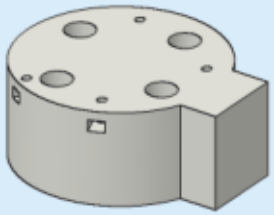
5.5 mm wrench for the M3 nuts.

Steps 1-12: wrist stack and housing with the servo

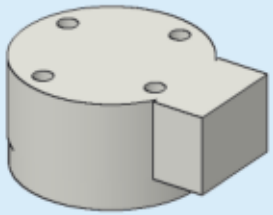
Steps 13-23: tower with the fingers (on the bench)

Steps 24-26: final assembly

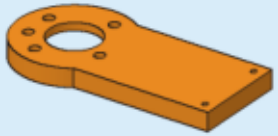
Parts inventory



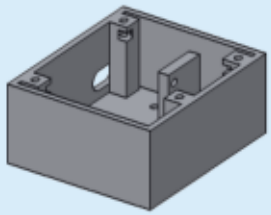
1x Spacer B



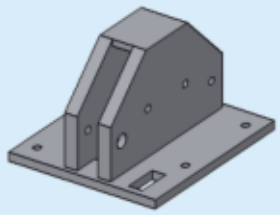
1x Spacer A



1x Camera mount



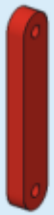
1x Housing box



1x Tower



1x Crank



1x Rod



1x OuterKnuckle_L



1x OuterKnuckle_R



2x OuterFinger



2x InnerKnuckle



2x InnerFinger



2x Fingertip

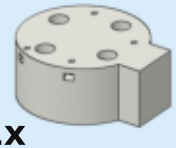
Hardware

6x M3 x 8	2x M3 x 45
9x M3 x 12	4x M6 x 75
9x M3 x 16	30x M3 nyloc nut
10x M3 x 30	50x M3 washer (7 x 0.5 mm)

STL: repository folder print/parts (the quantity is in the file name, _x2 = 2 pieces).

1

Nuts into spacer B

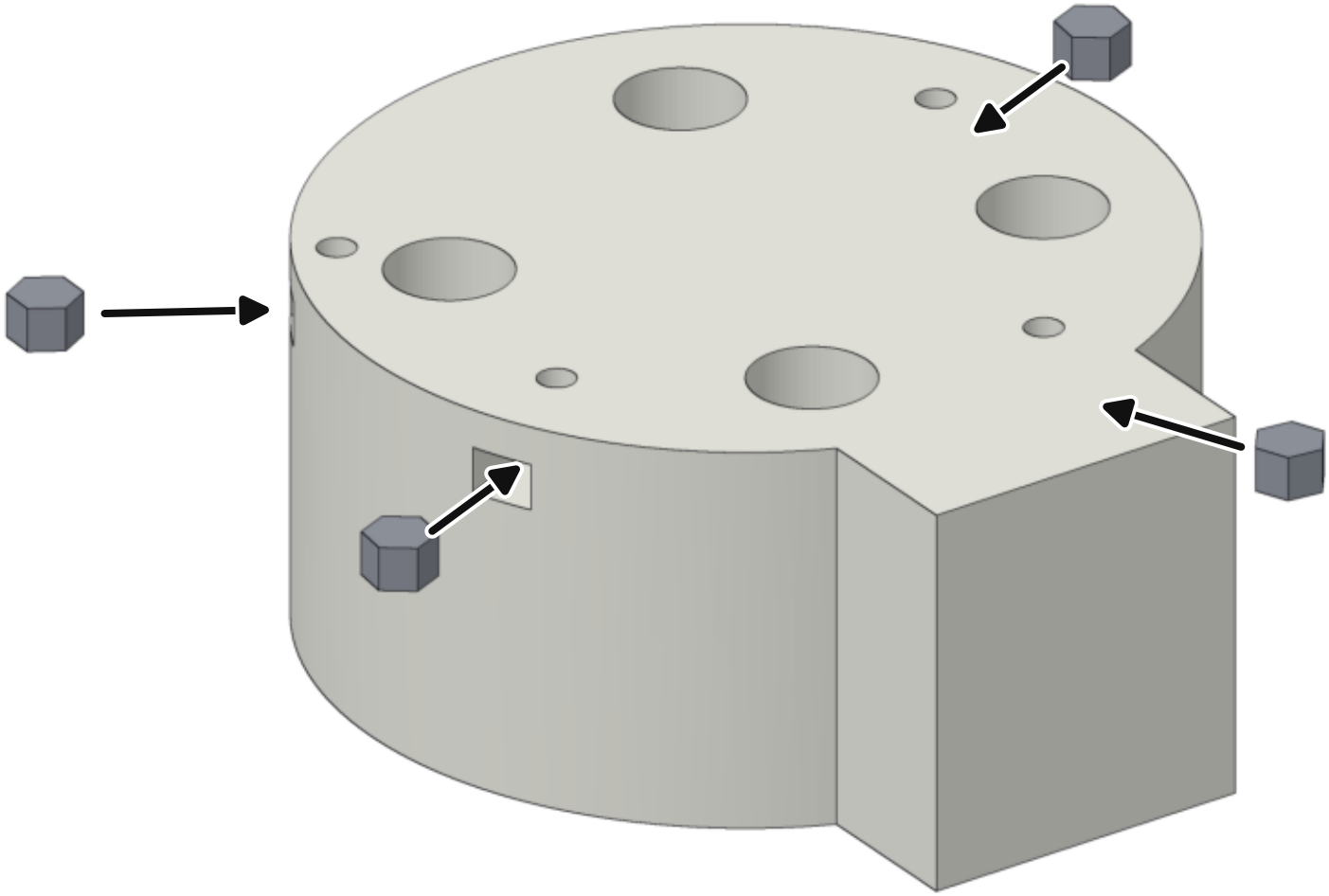


1x
Spacer B



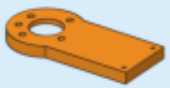
4x
M3 nyloc nut

- Push one nyloc nut into each side slot until it is centred on the hole.



2

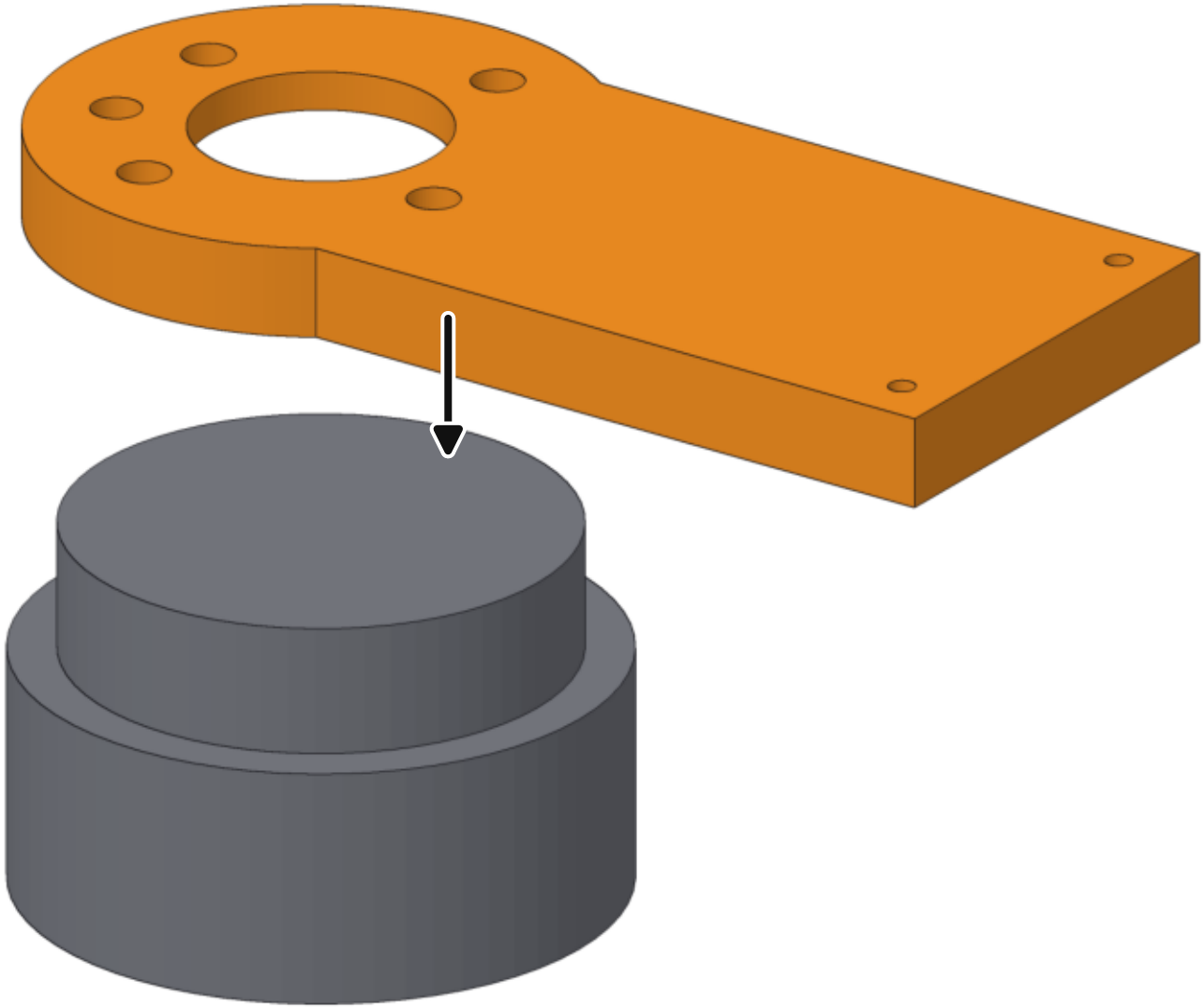
Camera mount onto the robot flange



1x
Camera mount

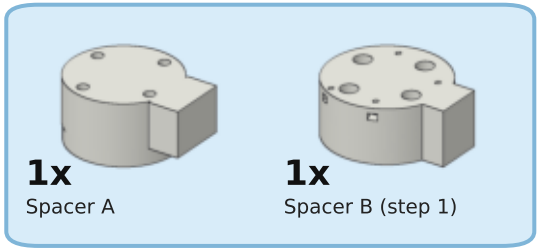
- The camera arm points to +X (the side where the red rod will be).

- Optional: 6 mm dowel pin in the centring hole.

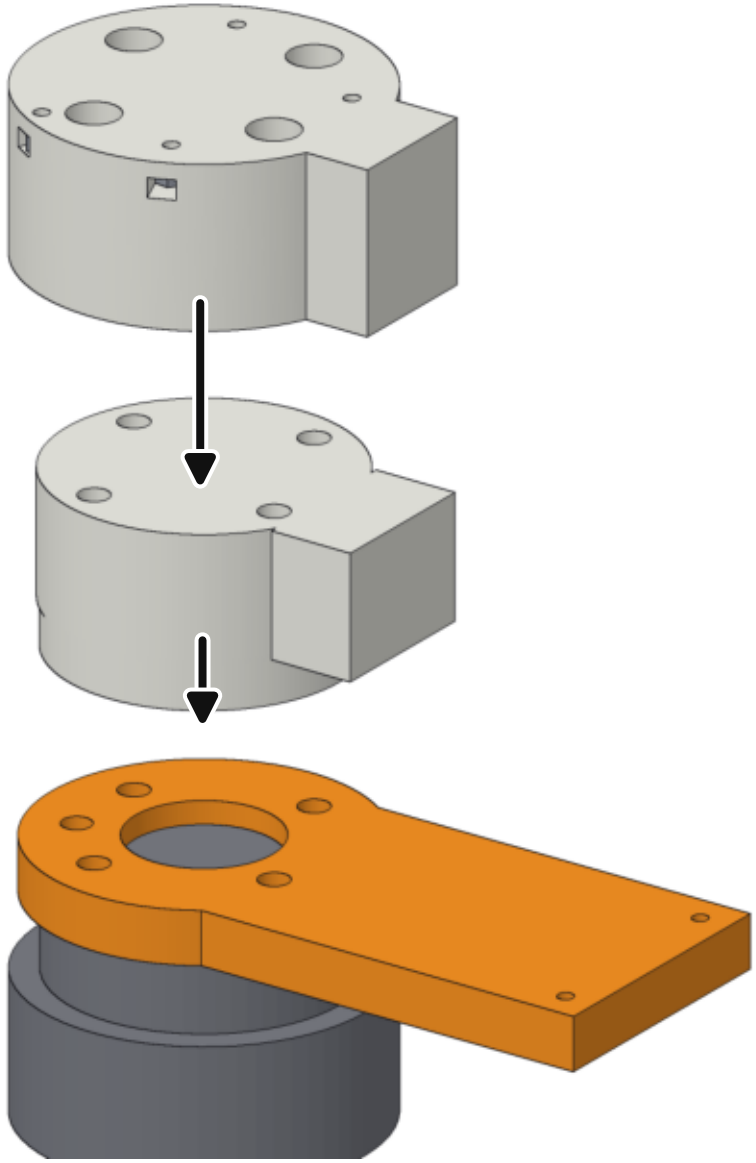


3

Spacers A and B



- The side block of both spacers faces the camera arm.



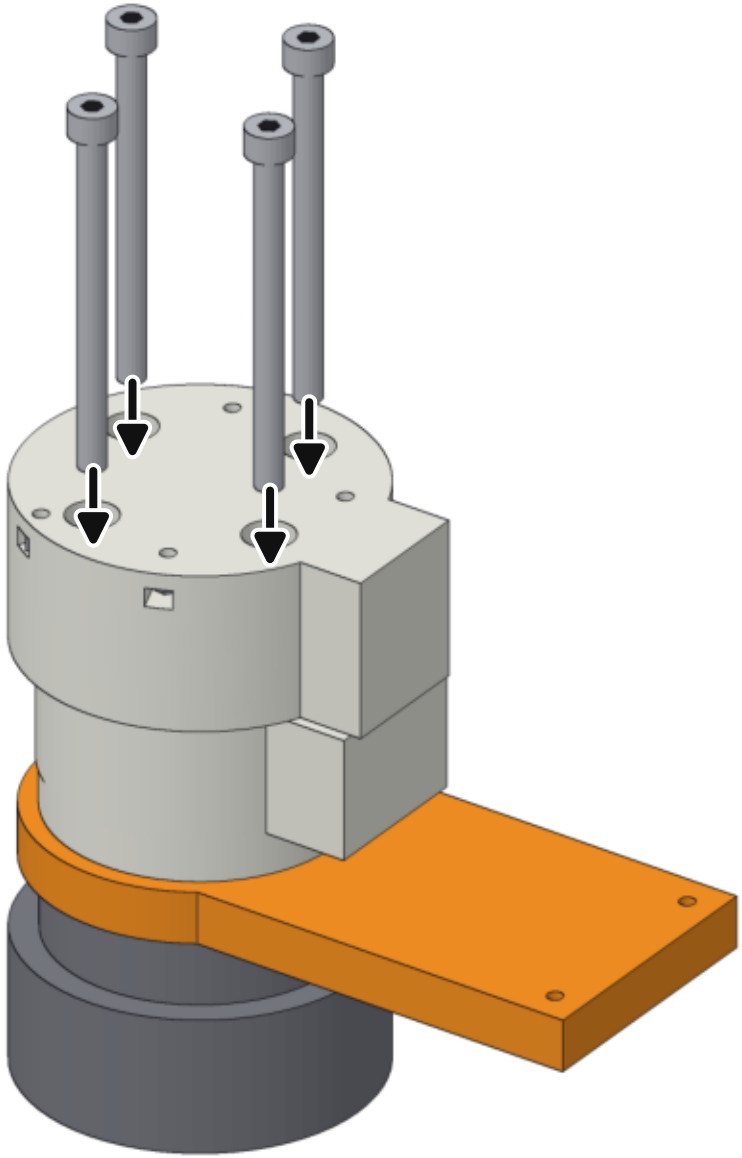
4

4 M6 bolts into the flange



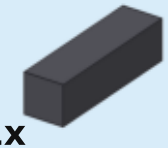
4x
M6 x 75

- Tighten crosswise. The heads sit recessed in spacer B.

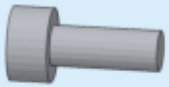


5

RealSense D435i camera

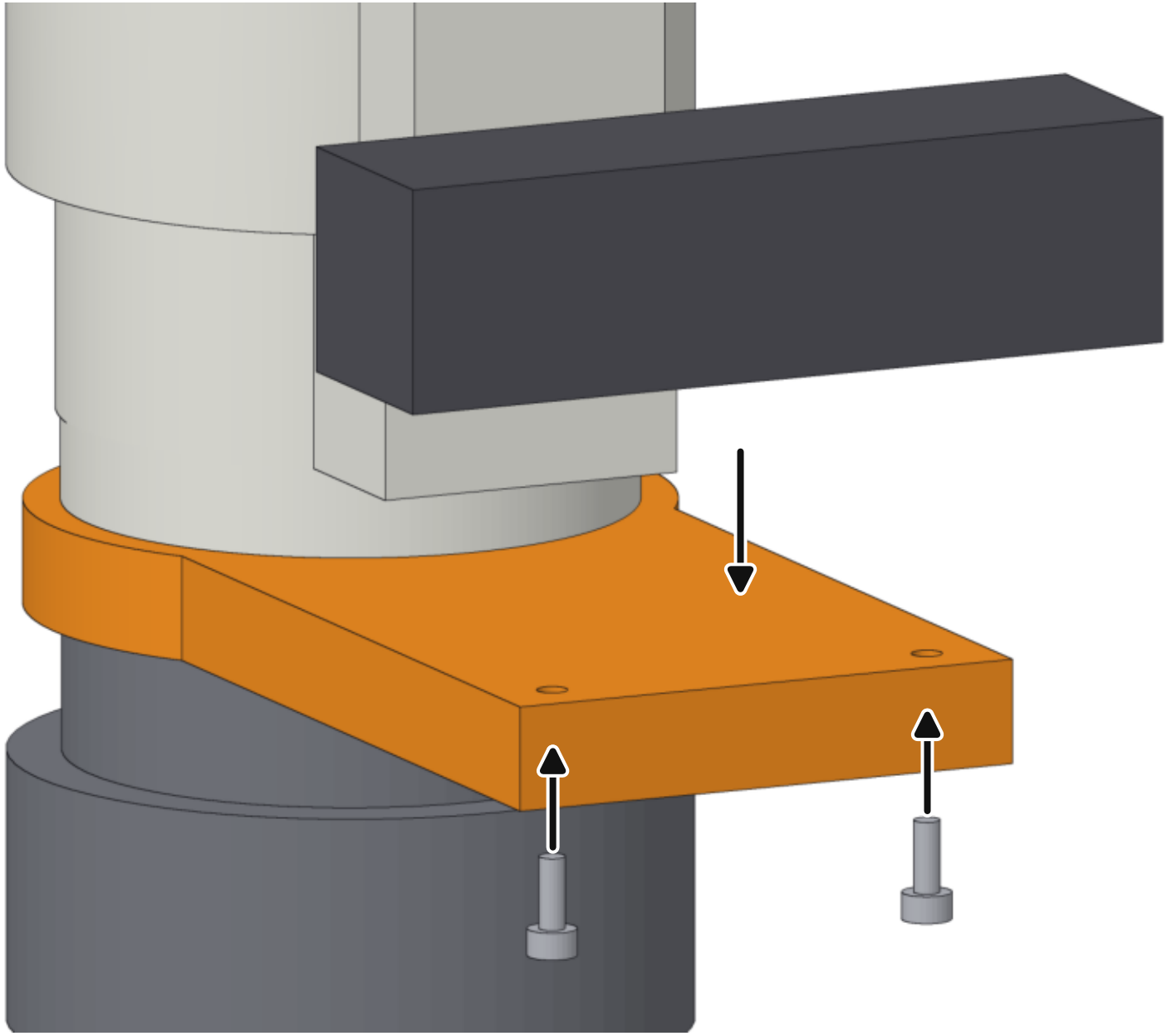


1x
RealSense D435i



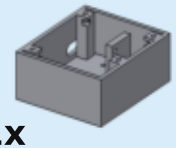
2x
M3 x 8

- Both screws go in from under the camera mount.



6

Nuts into the housing box

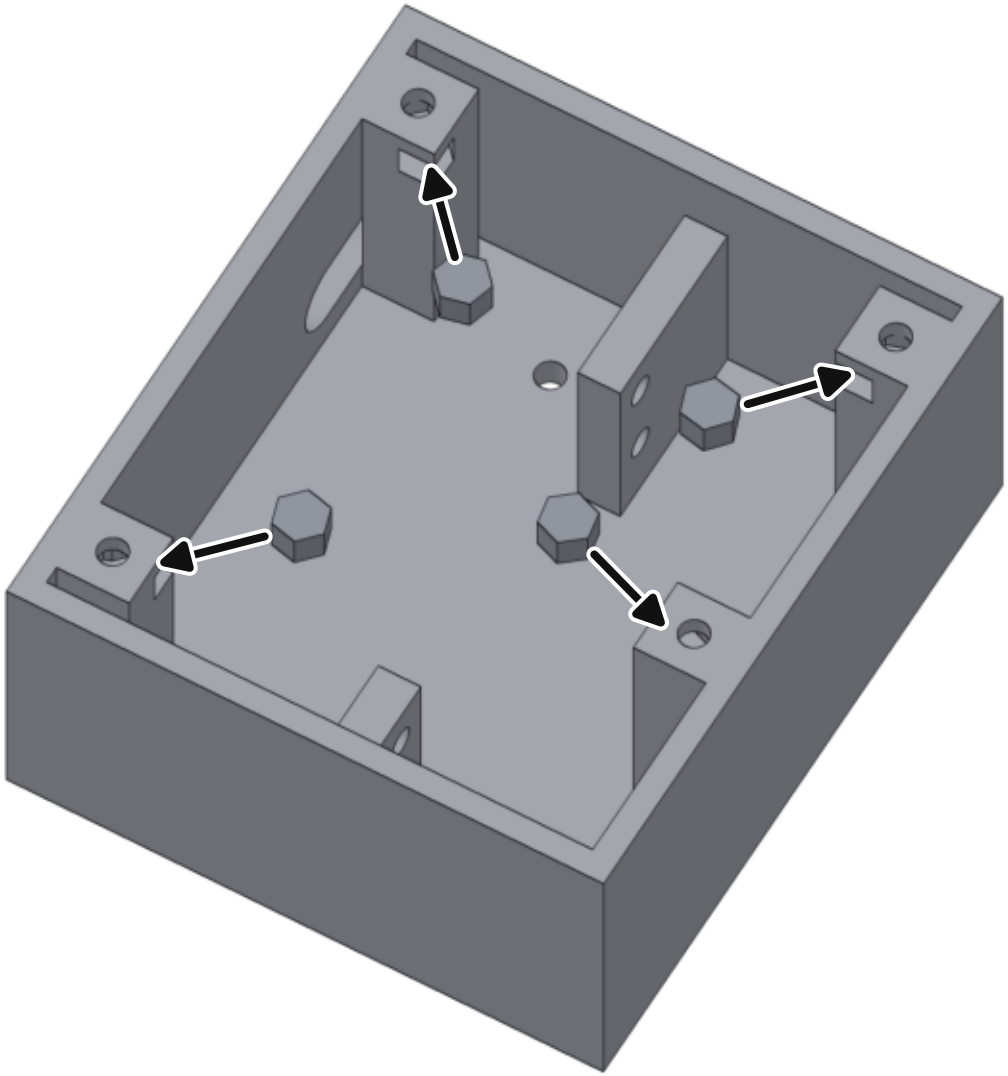


1x
Housing box



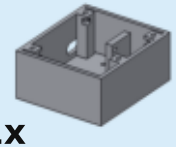
4x
M3 nyloc nut

- One nyloc nut in each corner column, through the inner slot.



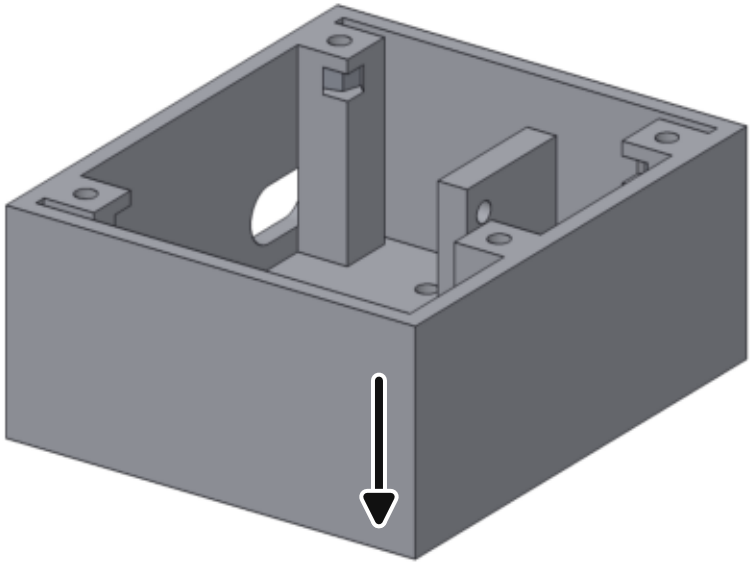
7

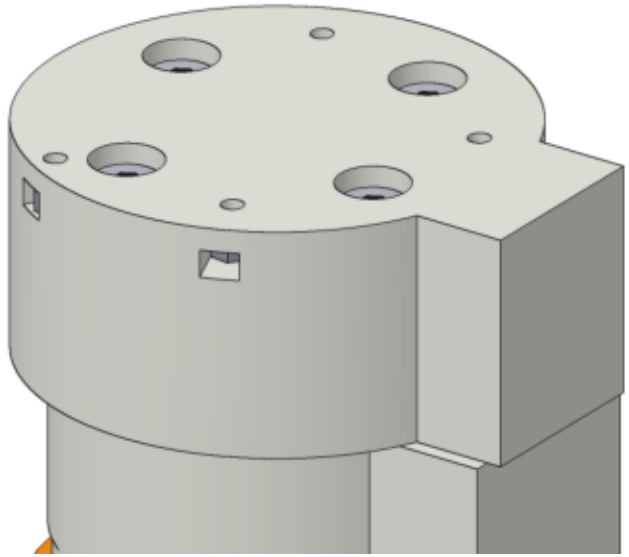
Housing box onto spacer B



1x
Housing box (step 6)

- The wall with the cable opening (-X) faces away from the camera.

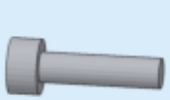




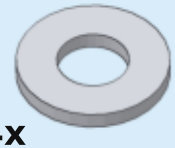


8

Housing box screws

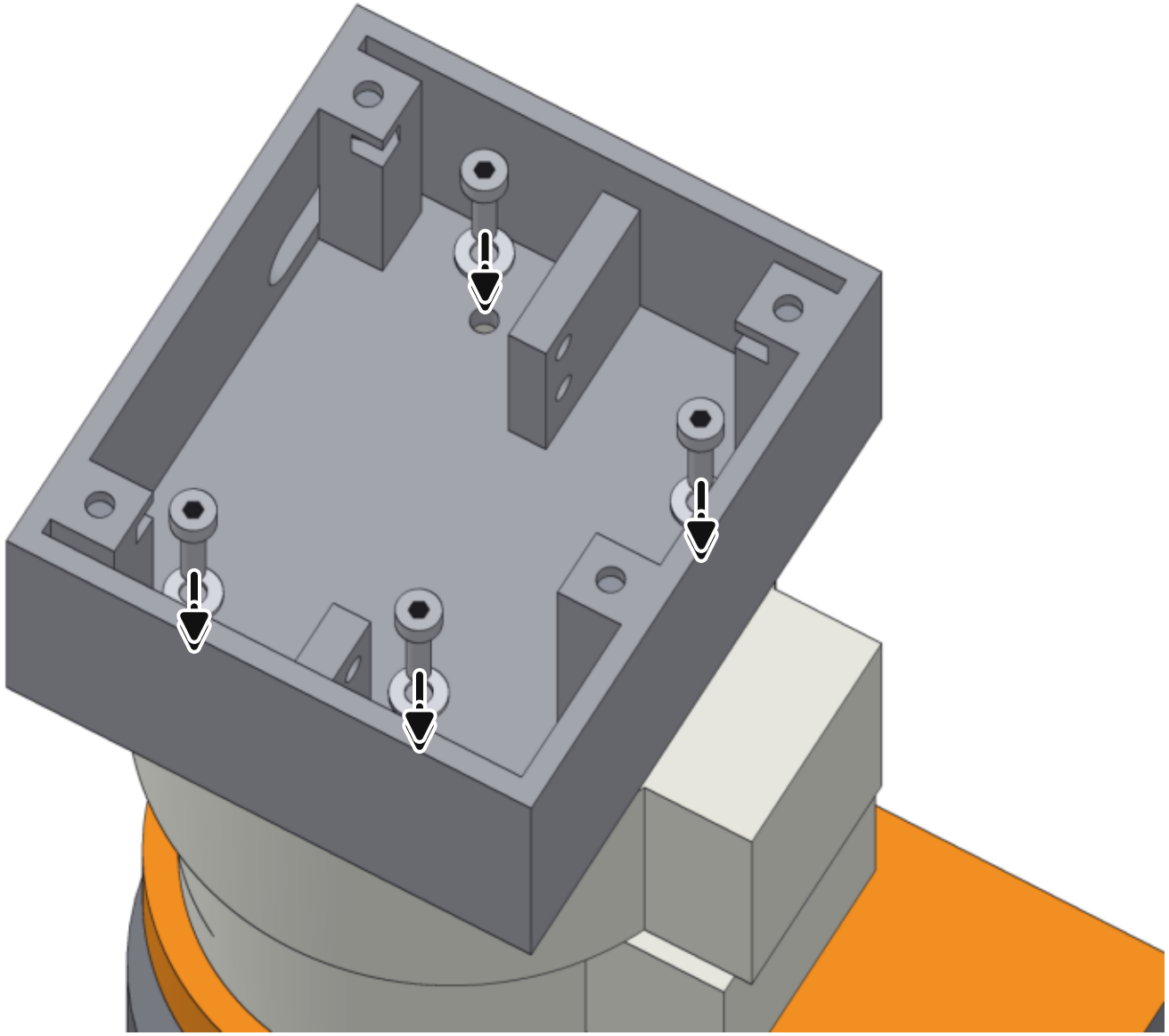


4x
M3 x 12



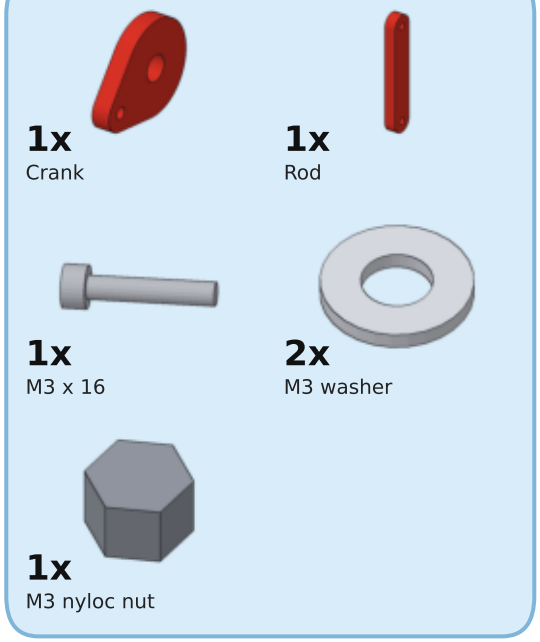
4x
M3 washer

- From inside the box, before fitting the servo.

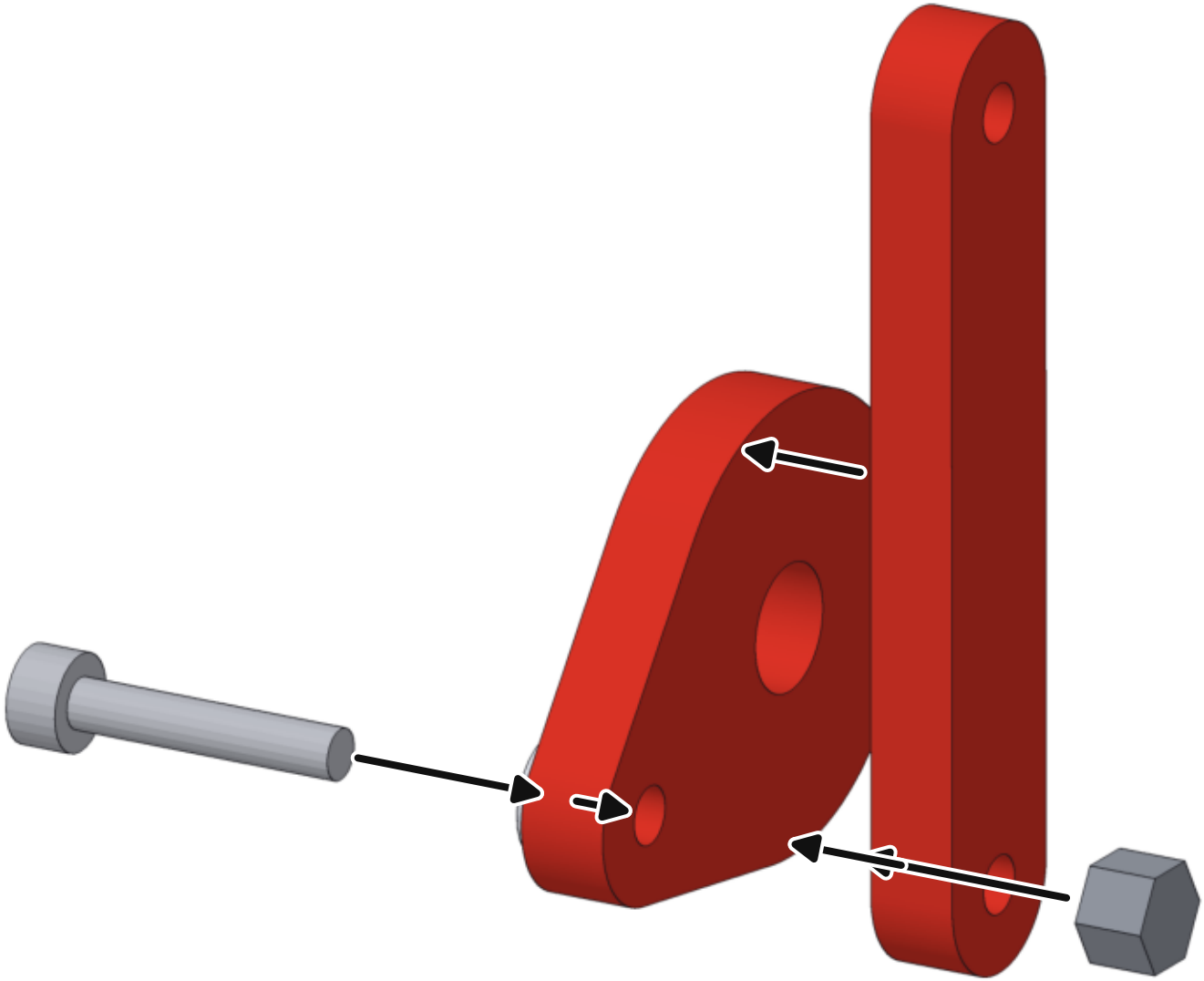


9

Crank + rod (on the bench)



- Tighten the nut only until the play is gone: the rod must turn freely.



10

Horn and crank onto the servo



1x
Servo D5545GS



1x
Metal horn

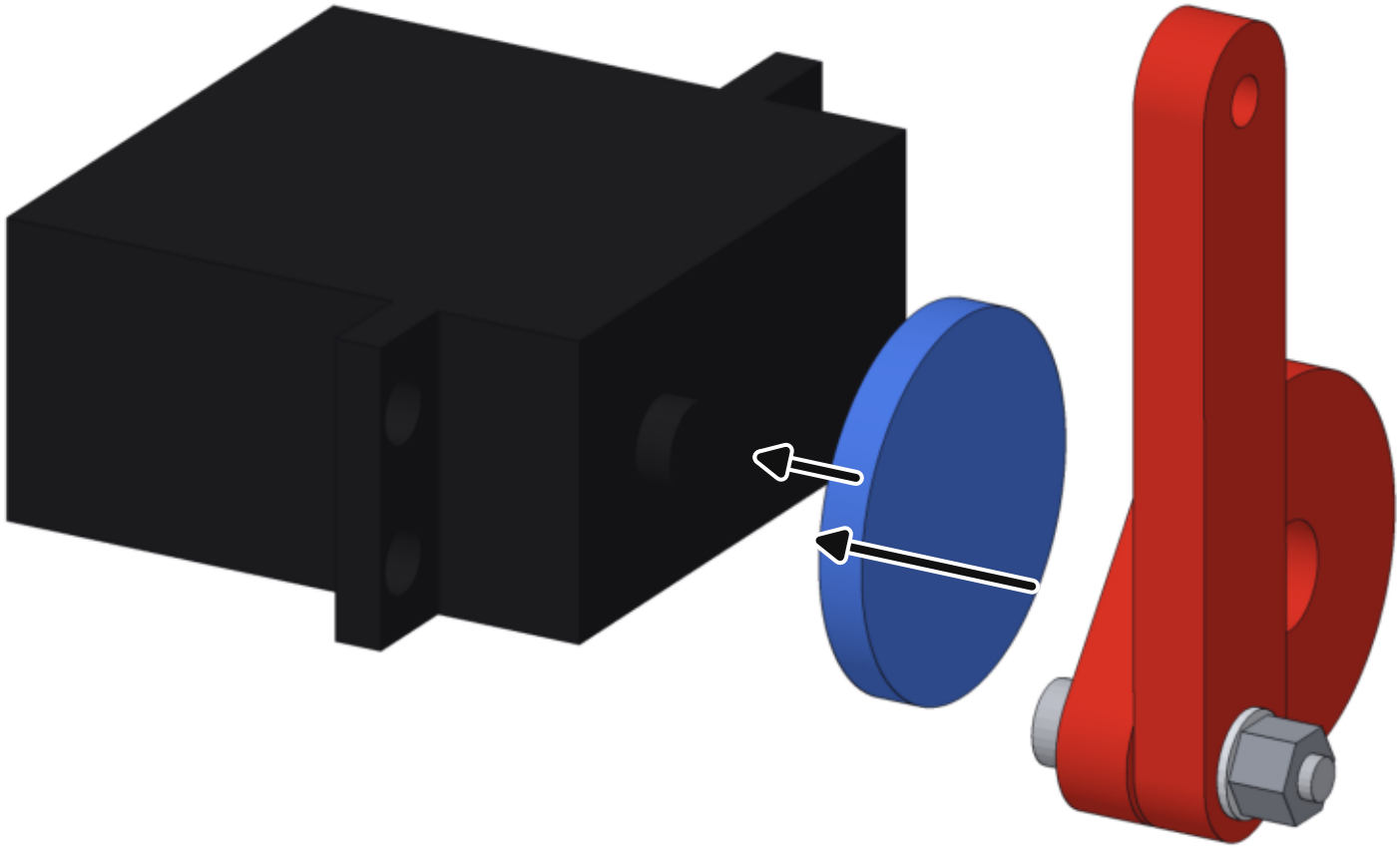


1x
Crank + rod (step 9)

- 1) Drill the crank using the metal horn as a template.

- 2) Move the servo to its centre (1500 us).

- 3) Fit the horn with the crank pointing to -Y (as drawn) and the rod vertical.



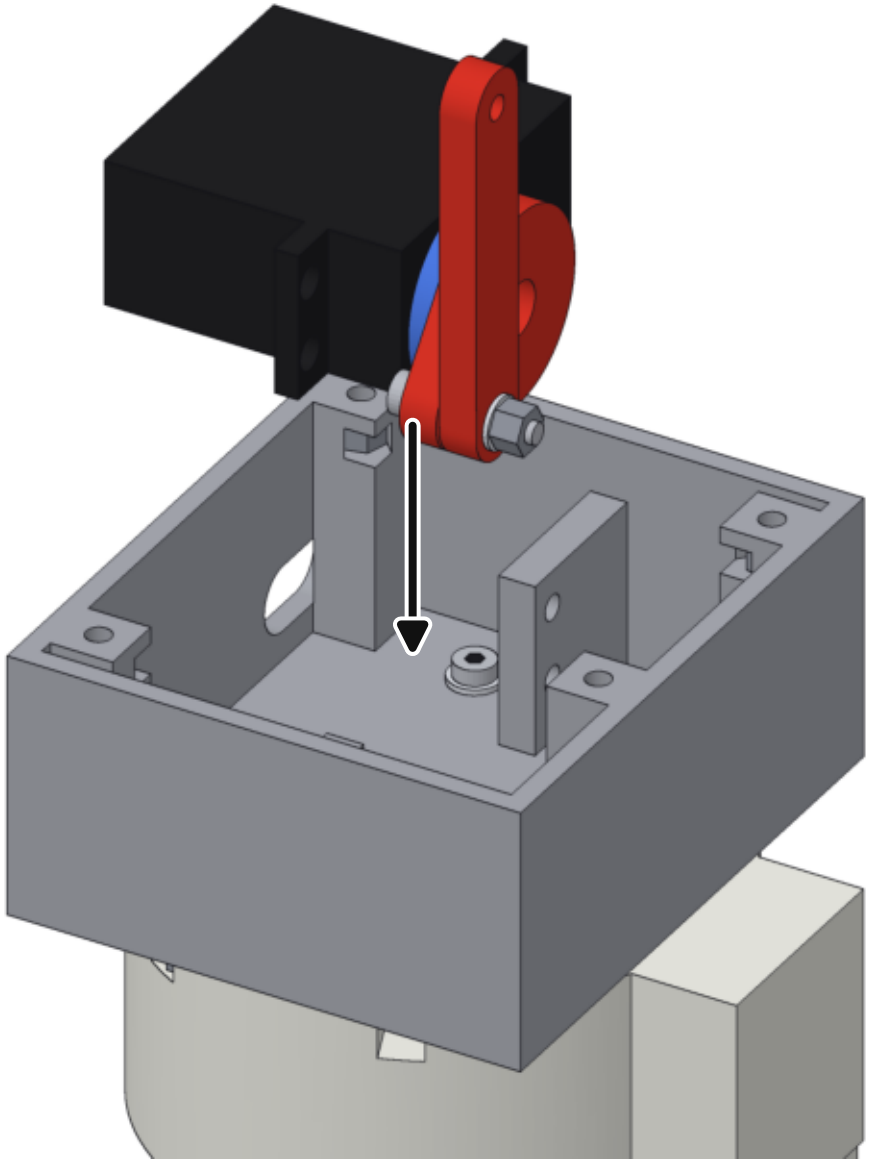
11

Servo into the housing box



1x
Servo + crank (step 10)

- Lower the servo from above: its ears rest on the inner bulkhead.

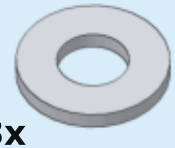


12

Servo screws



4x
M3 x 16

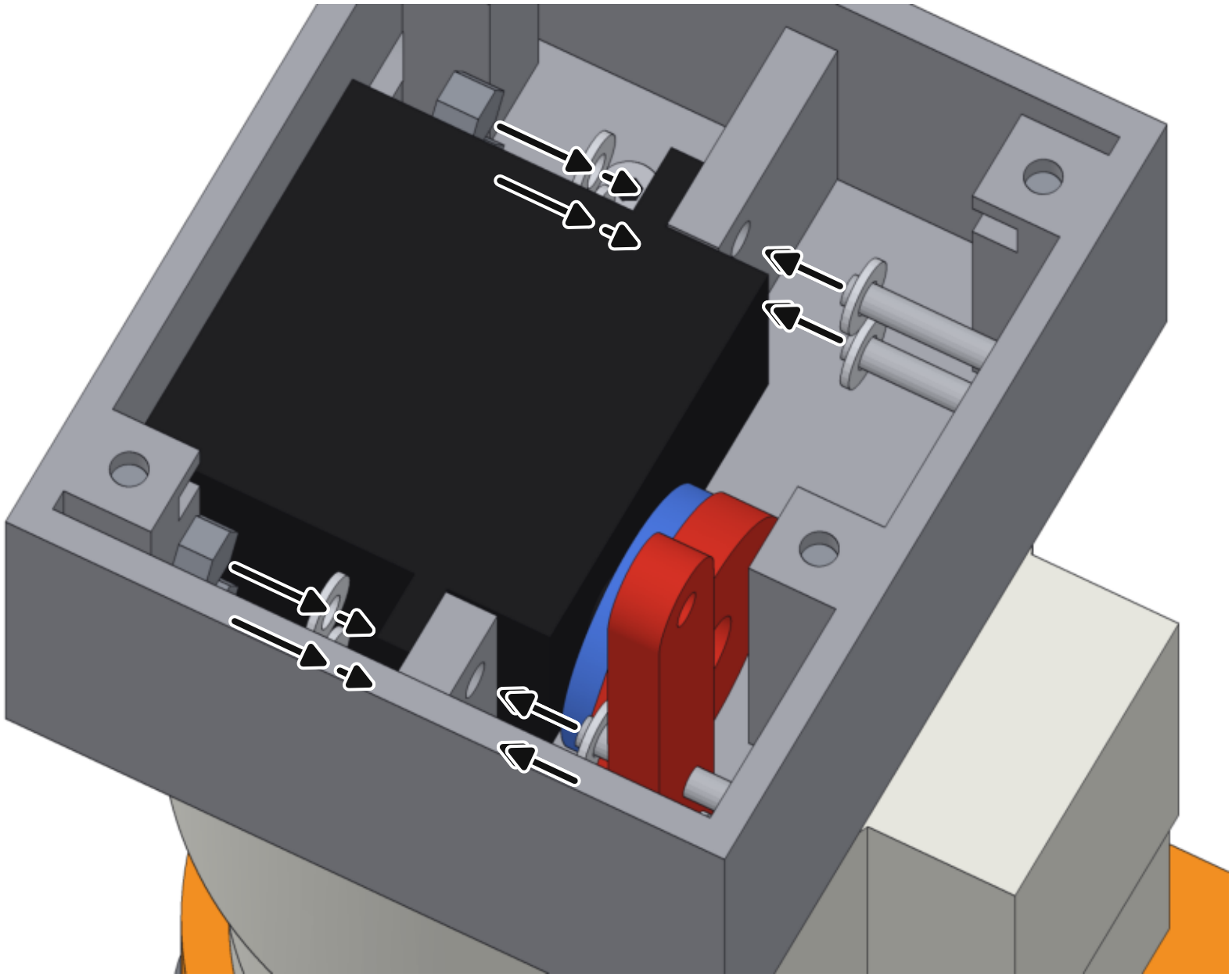


8x
M3 washer



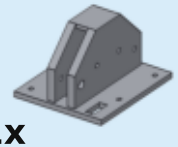
4x
M3 nyloc nut

- Head on the bulkhead side (+X), nut on the servo-ear side.



13

Outer knuckles into the tower (on the bench)



1x
Tower



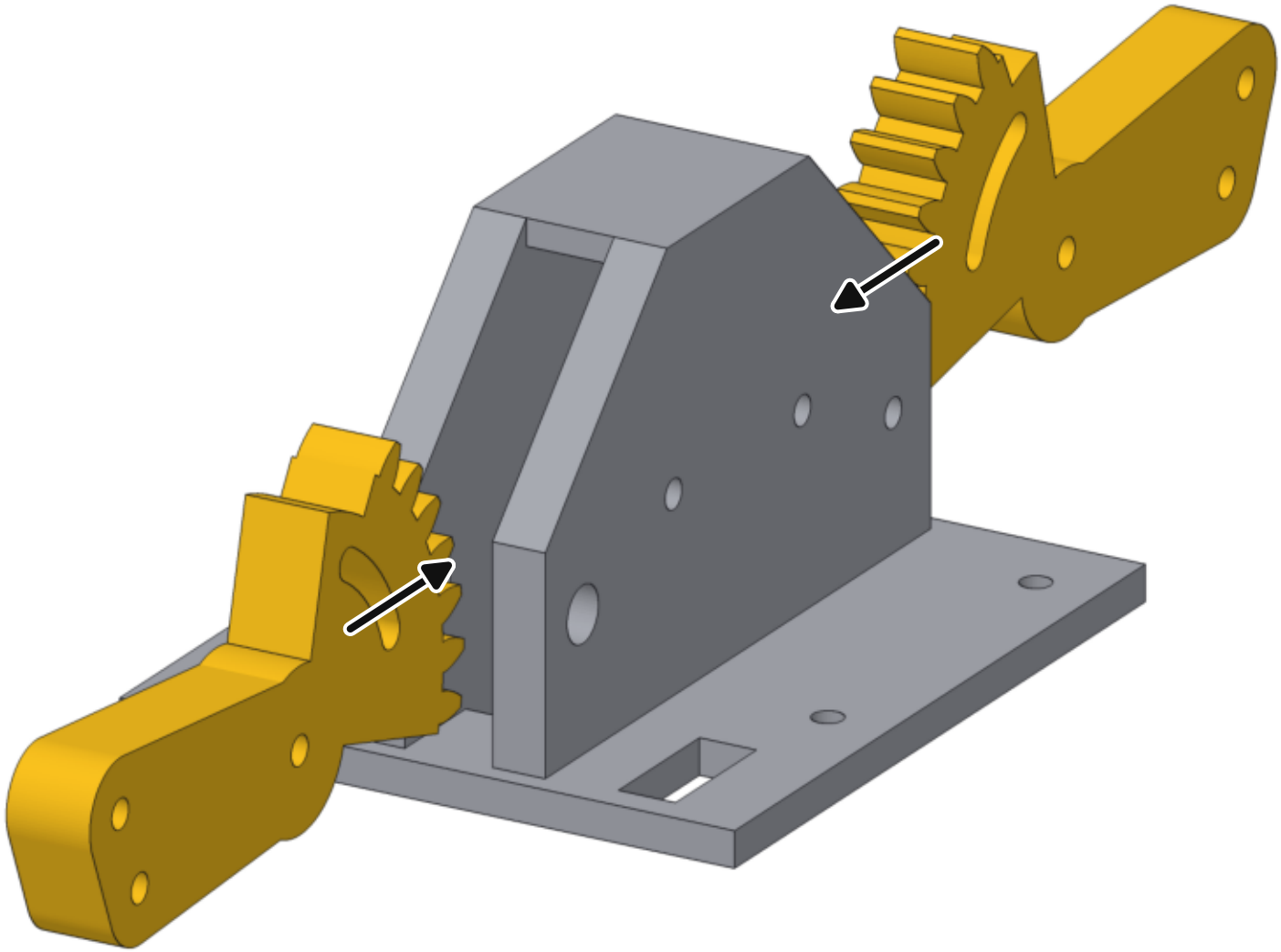
1x
OuterKnuckle_L



1x
OuterKnuckle_R

- OuterKnuckle_L goes on the -Y side (rod side). L and R are NOT interchangeable.

- Mesh both with the fingers open (symmetric).

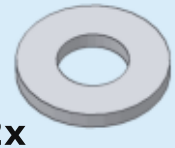


14

Outer knuckle pivots



2x
M3 x 30

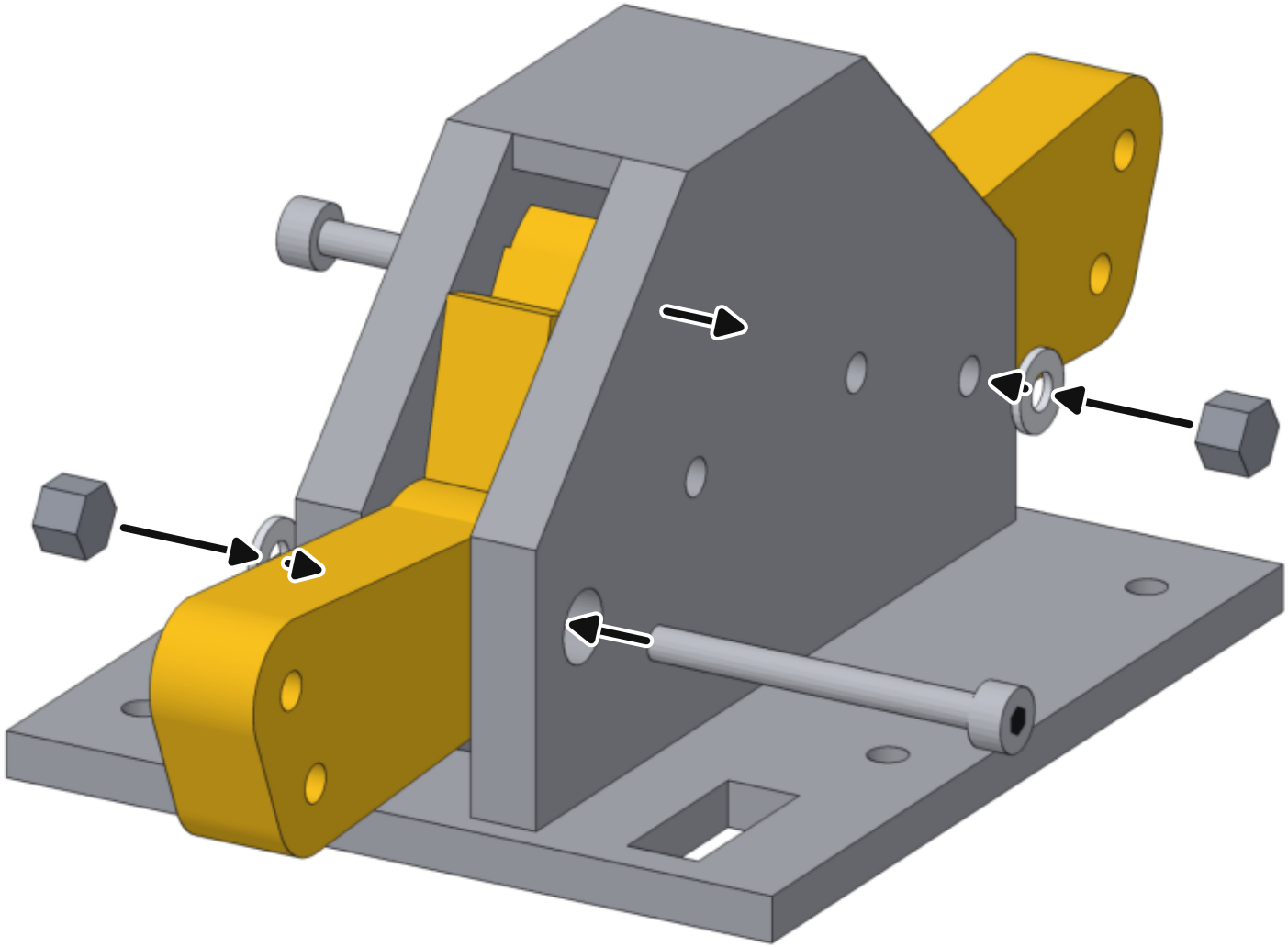


2x
M3 washer



2x
M3 nyloc nut

- Left: head recessed in the +X face. Right: head recessed in the -X face.



15

Nut into the ear (on the bench)

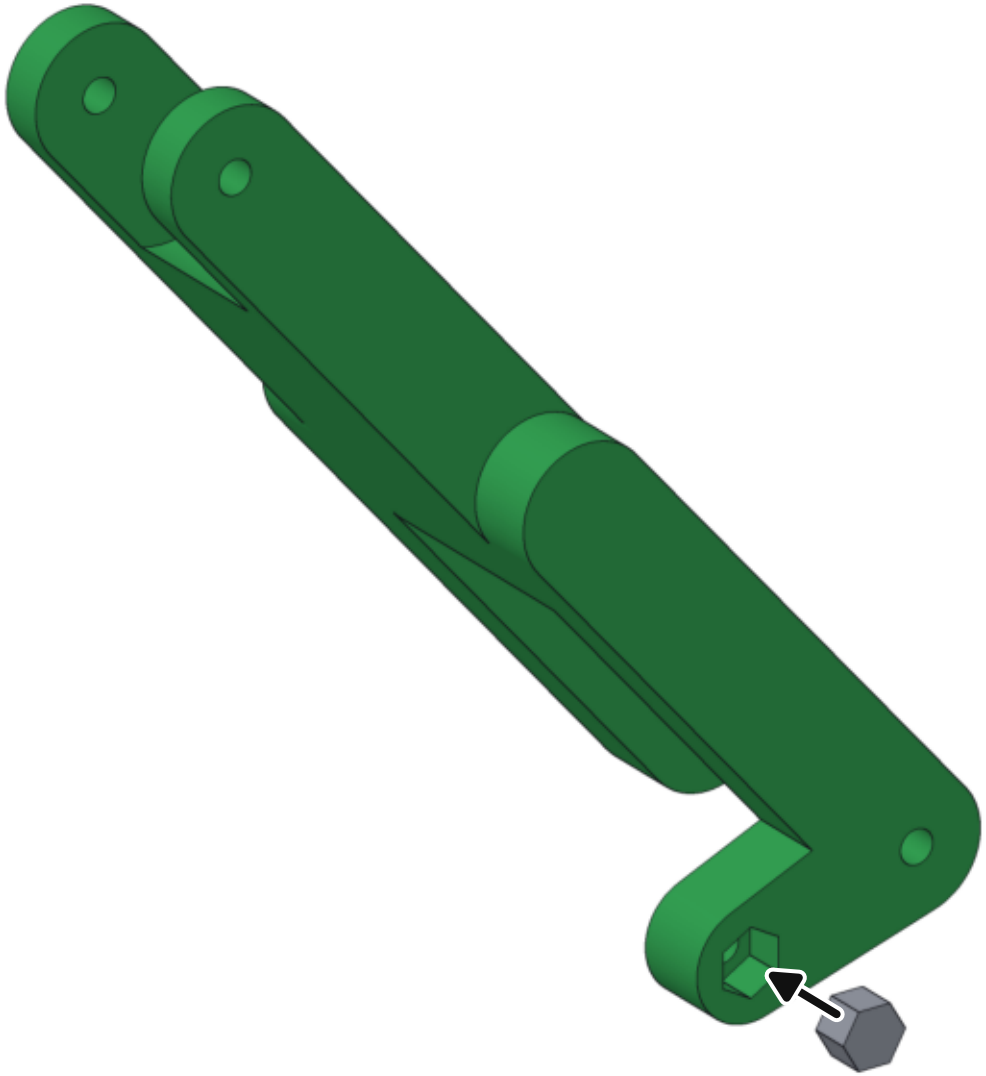


1x
InnerKnuckle



1x
M3 nyloc nut

- Only the inner knuckle on the -Y side uses its ear. Push the nut to the bottom of the hex pocket.



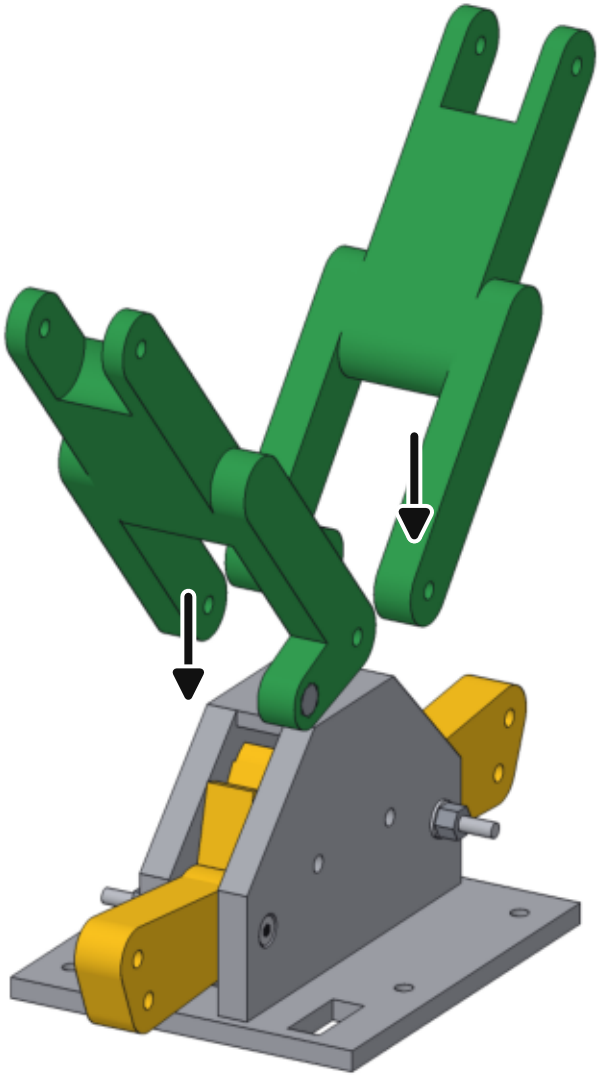
16

Inner knuckles



2x
InnerKnuckle

- The prongs straddle the tower. The ear with the nut ends up on the -Y side, +X face.

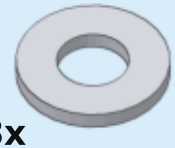


17

Inner knuckle pivots



2x
M3 x 45

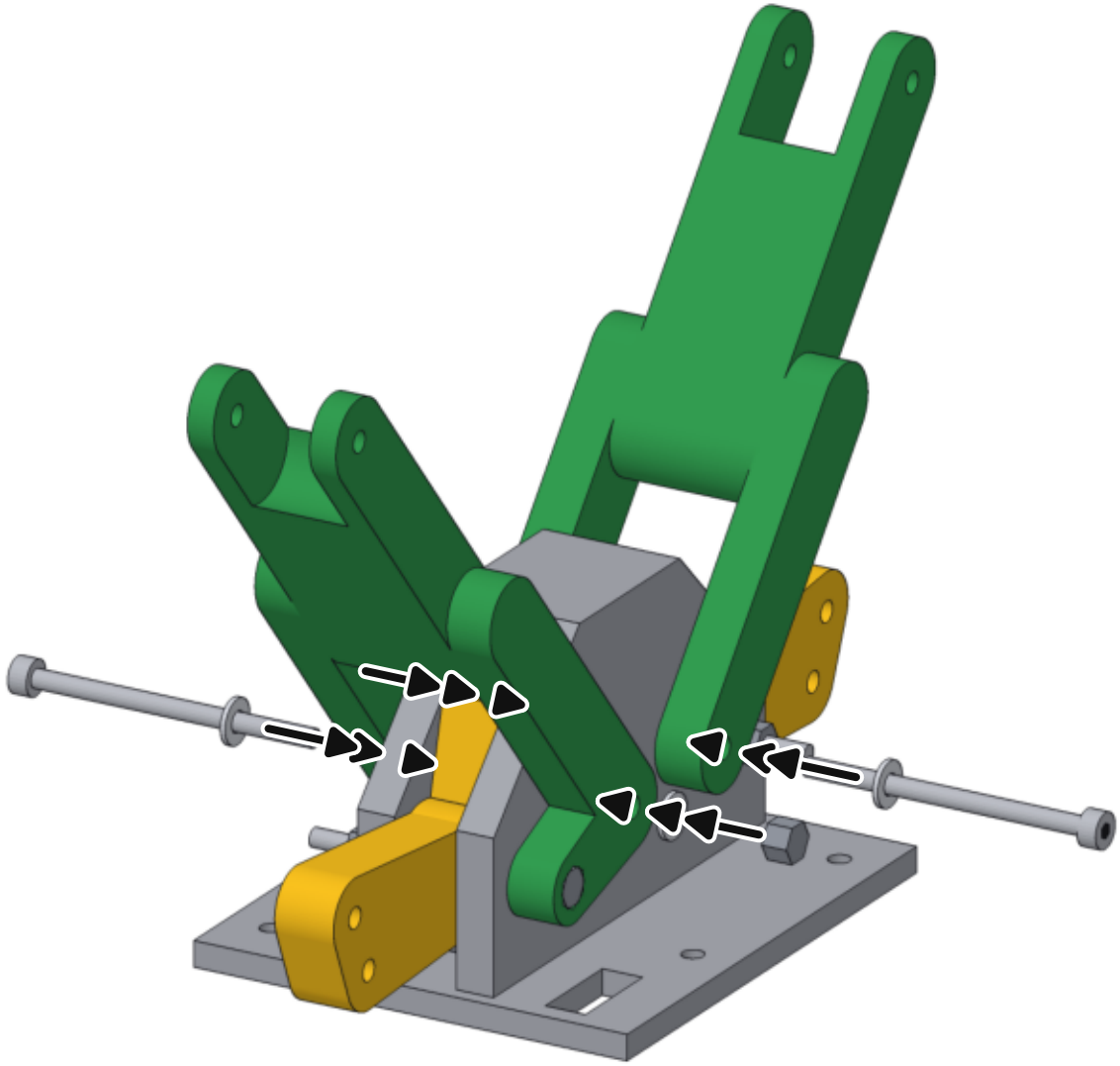


8x
M3 washer



2x
M3 nyloc nut

- Order: prong - washer - wall - gear slot - wall - washer - prong.



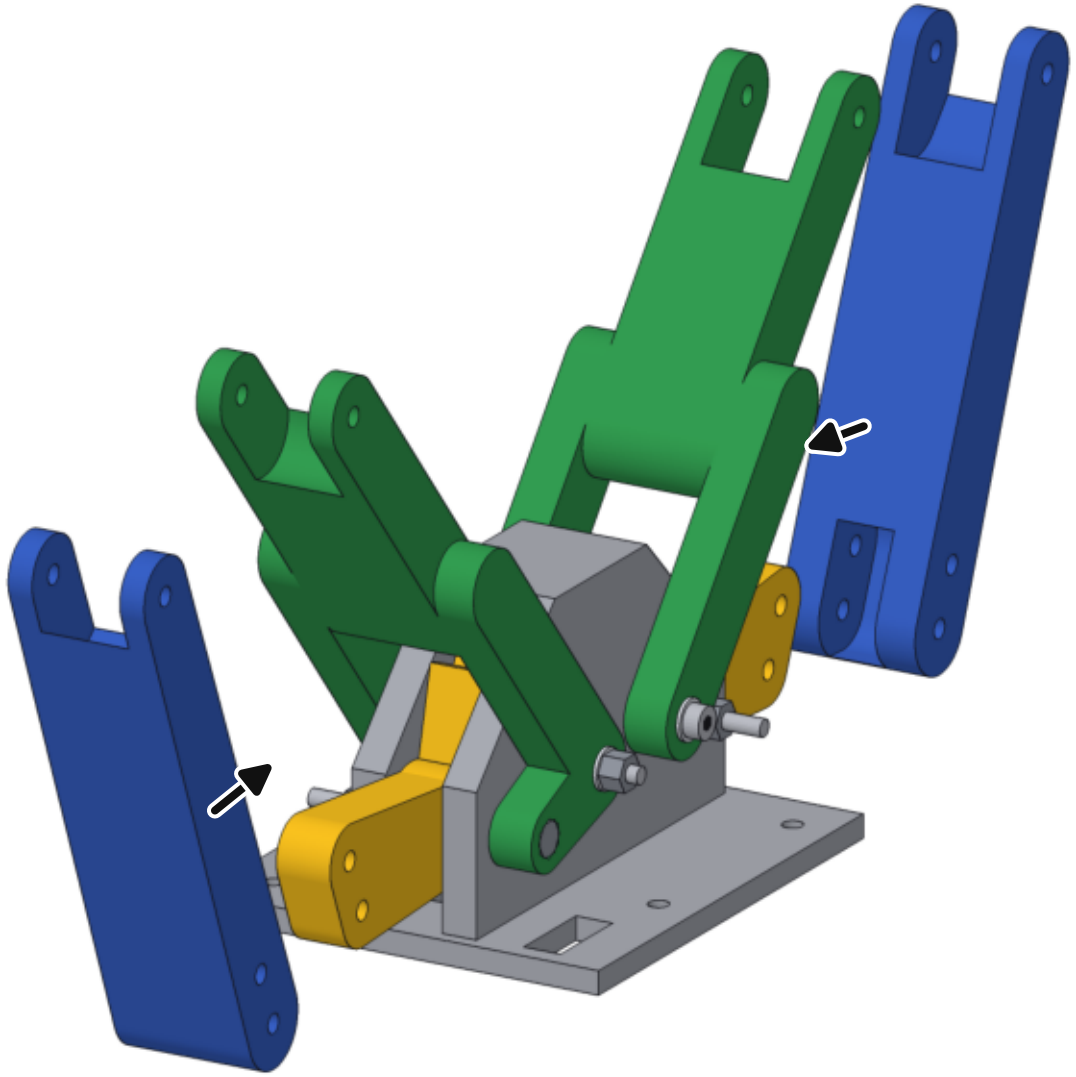
18

Outer fingers



2x
OuterFinger

- The outer-finger fork straddles the yellow arm.

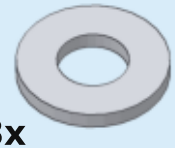


19

Outer knuckle - outer finger joint



4x
M3 x 30

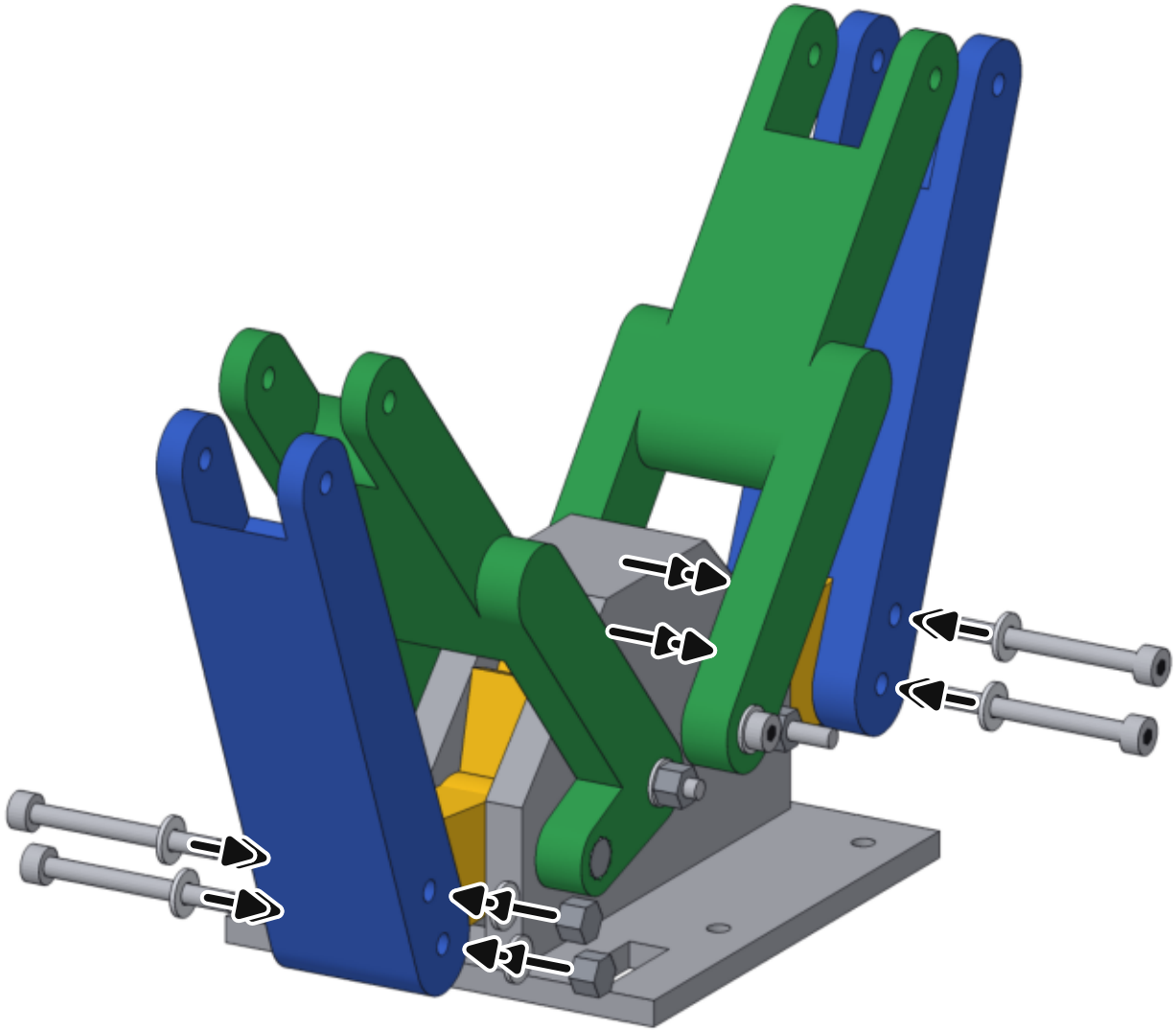


8x
M3 washer



4x
M3 nyloc nut

- 2 bolts per side: this joint is rigid, tighten firmly.



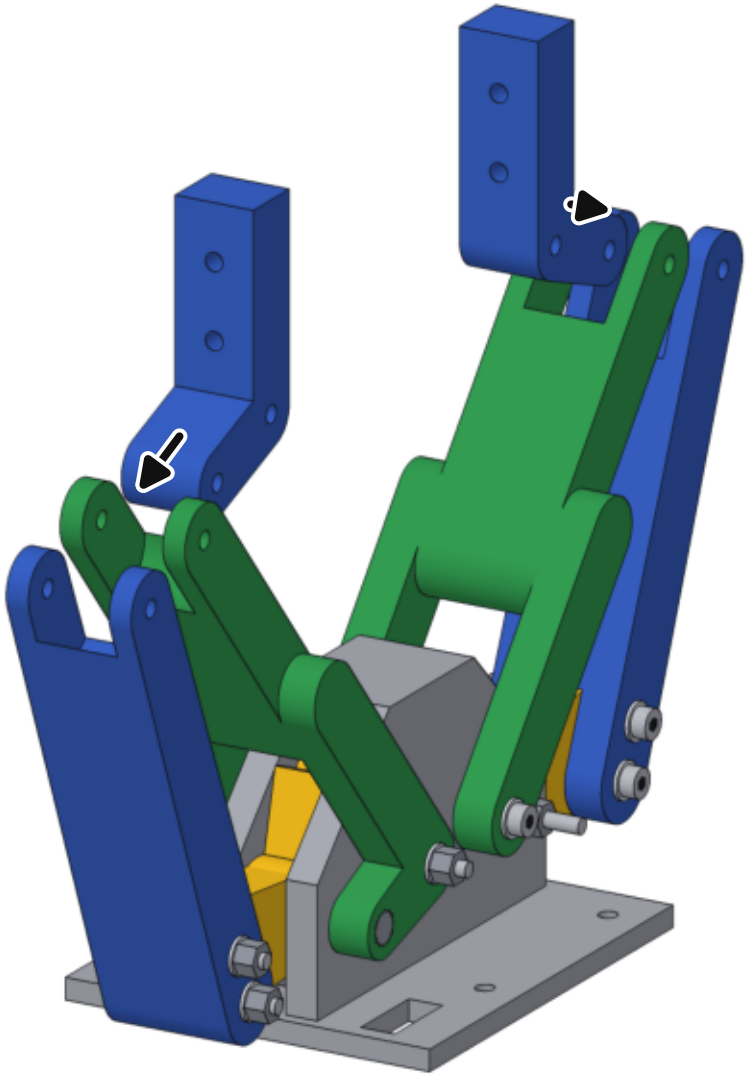
20

Inner fingers



2x
InnerFinger

- They slide into the forks of the outer finger (blue) and the inner knuckle (green).

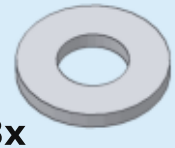


21

Inner finger pivots



4x
M3 x 30

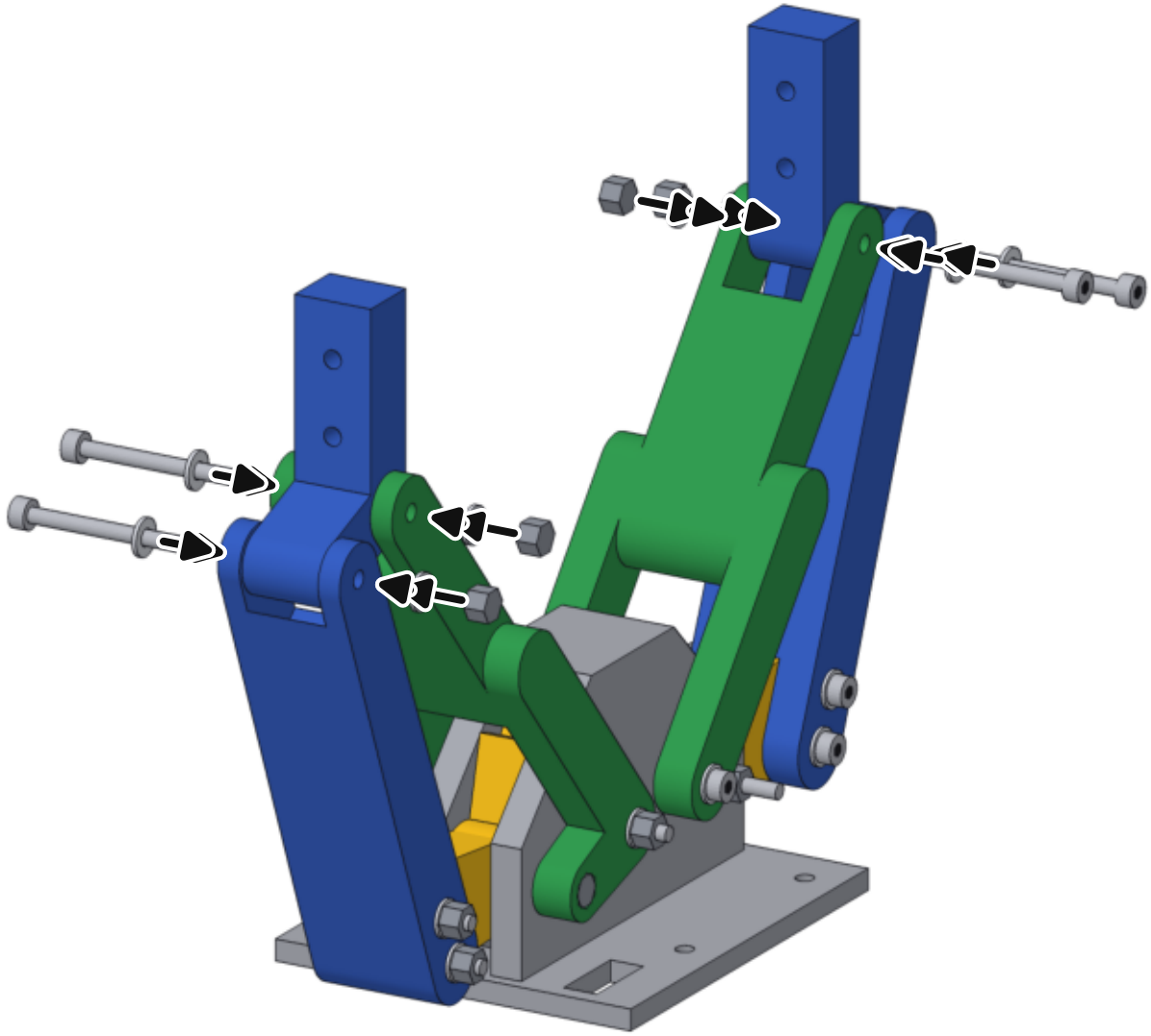


8x
M3 washer



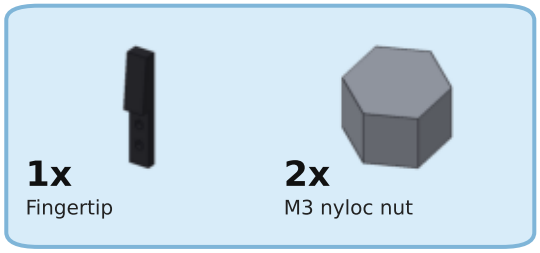
4x
M3 nyloc nut

- Tighten until the play is gone; the fingers must move smoothly.



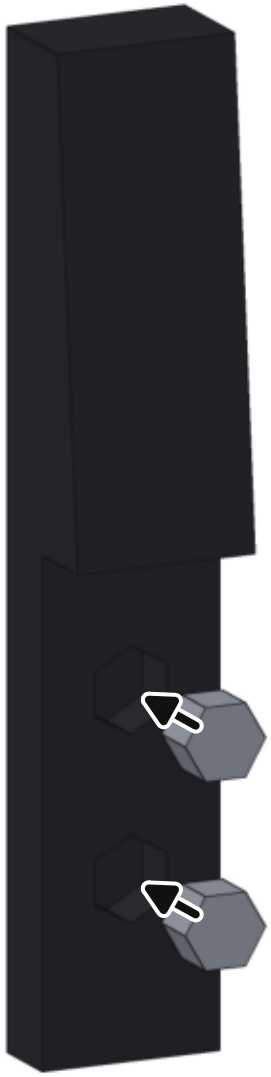
22

Nuts into the fingertips (on the bench)



- 2 nuts in the hex pockets on the back of each fingertip.

- Repeat with the other fingertip.



x2

23

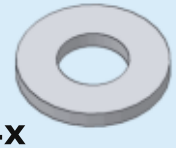
Fingertips onto the inner fingers



2x
Fingertip

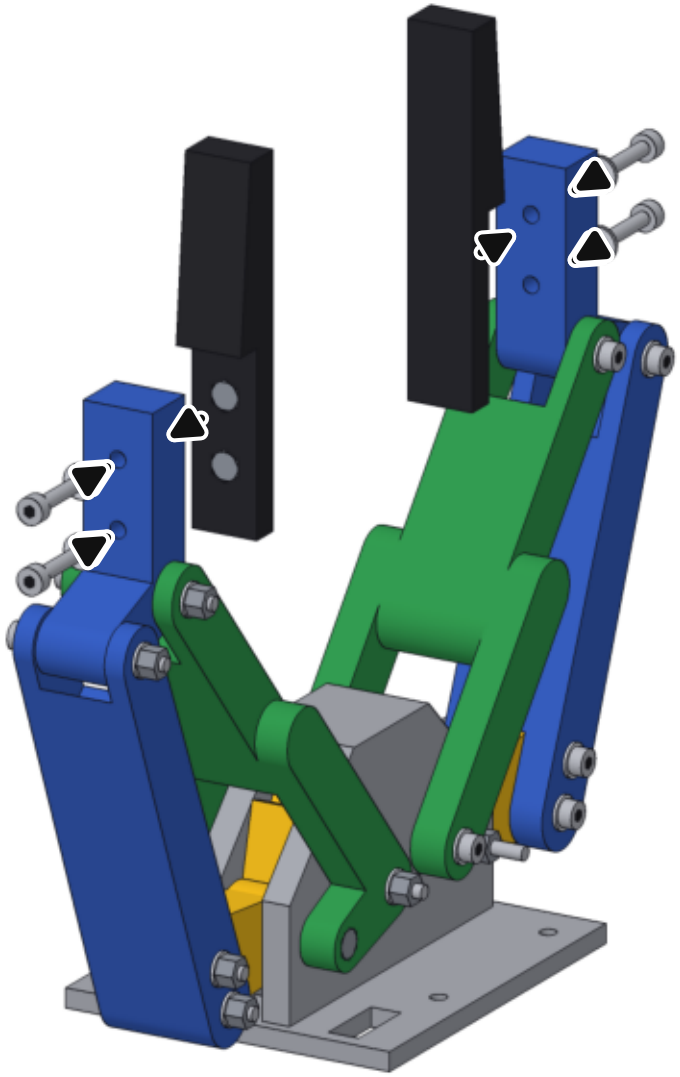


4x
M3 x 16



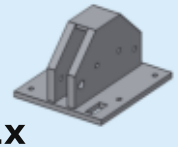
4x
M3 washer

- Screws go in from the outer face of the inner finger.



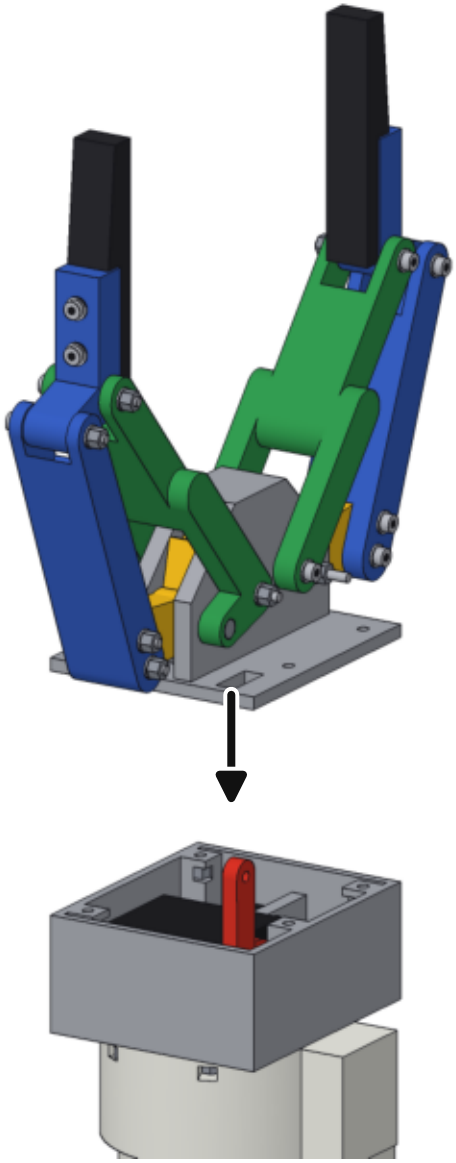
24

Tower onto the housing box



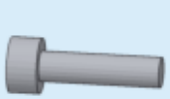
1x
Tower with fingers (steps 13-23)

- Pass the red rod through the slot in the tower plate.

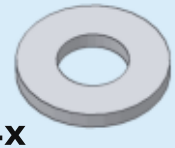


25

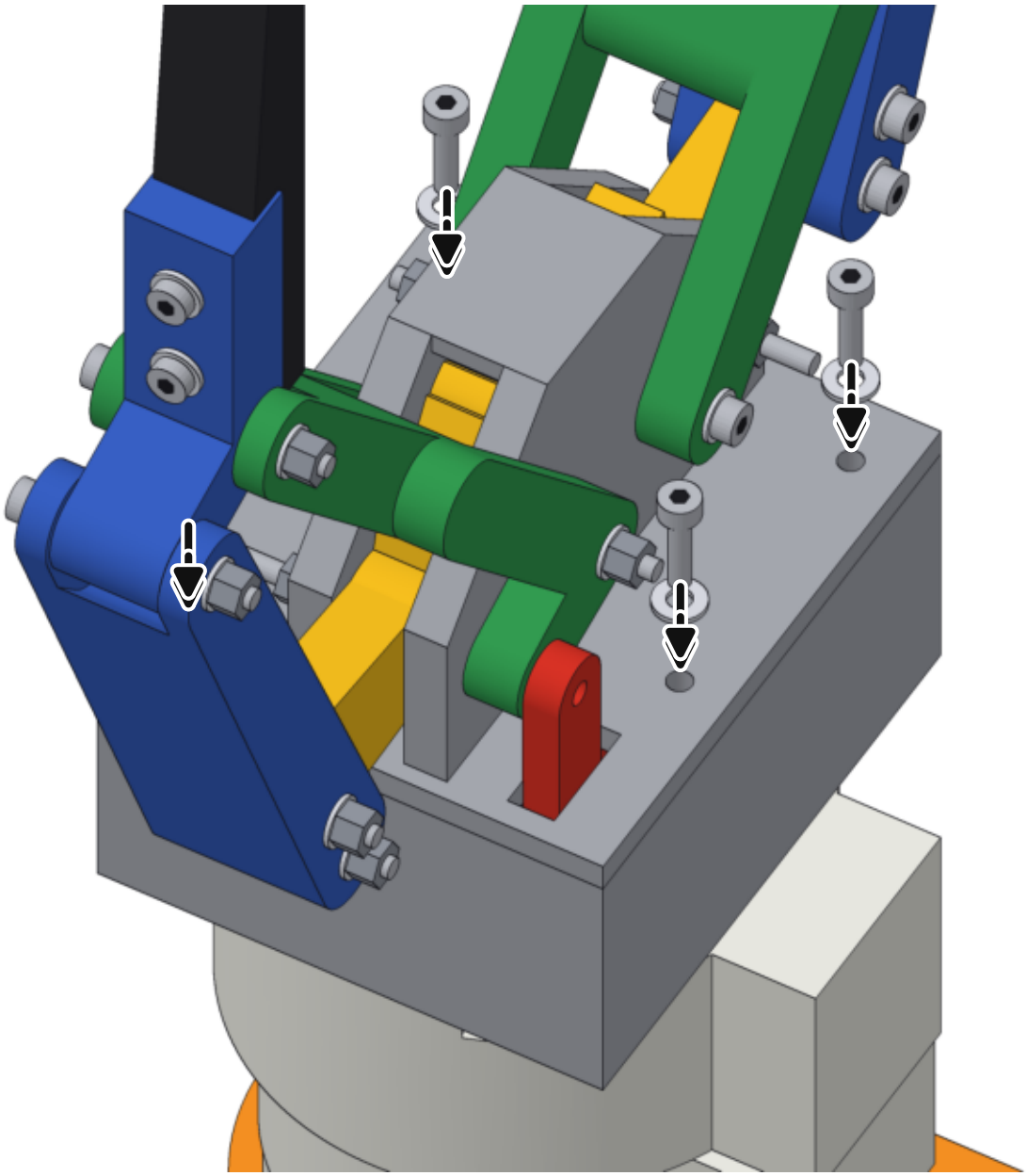
Tower screws



4x
M3 x 12

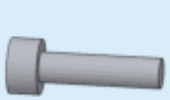


4x
M3 washer

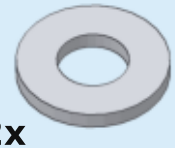


26

Rod to the ear



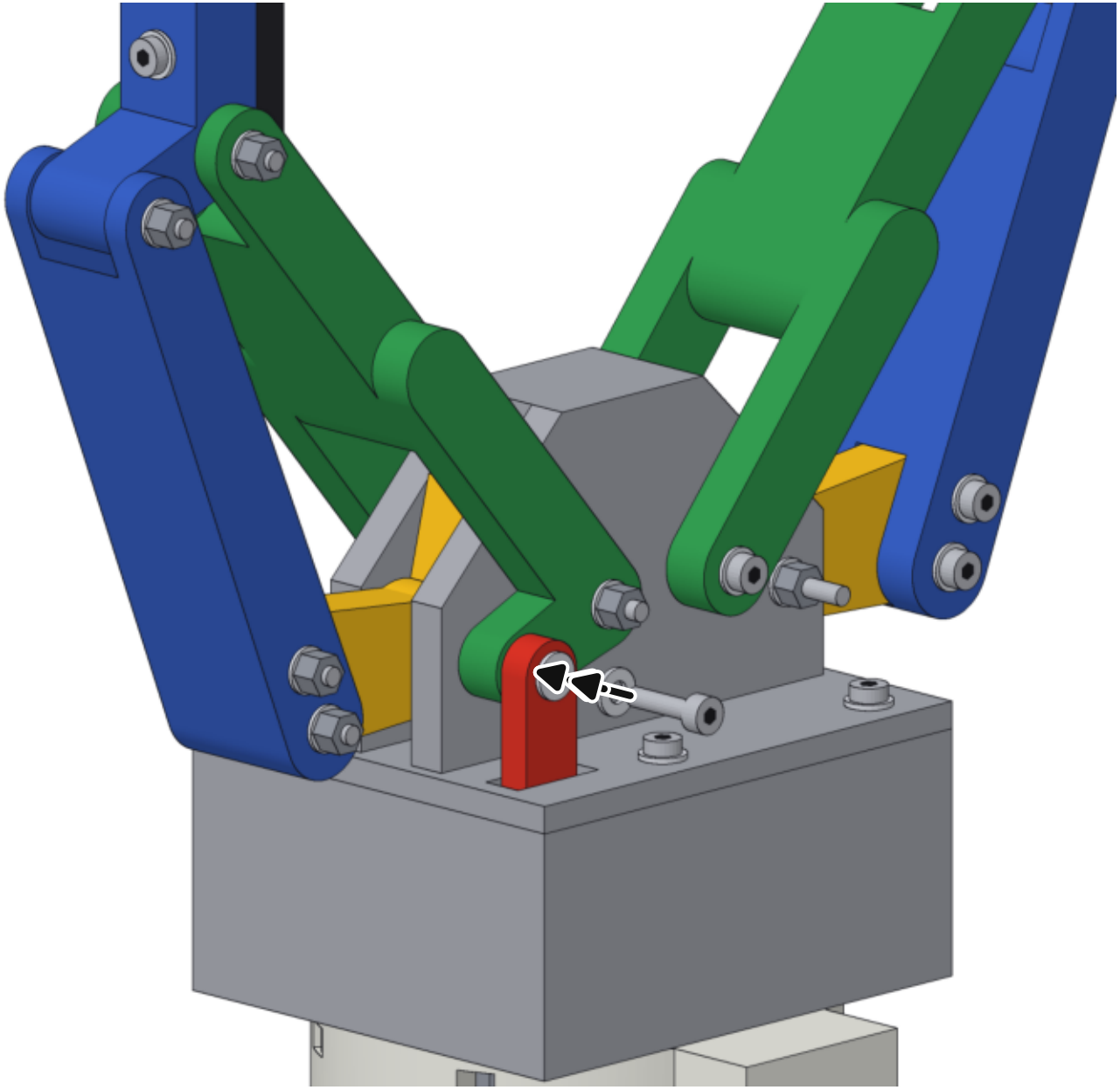
1x
M3 x 12



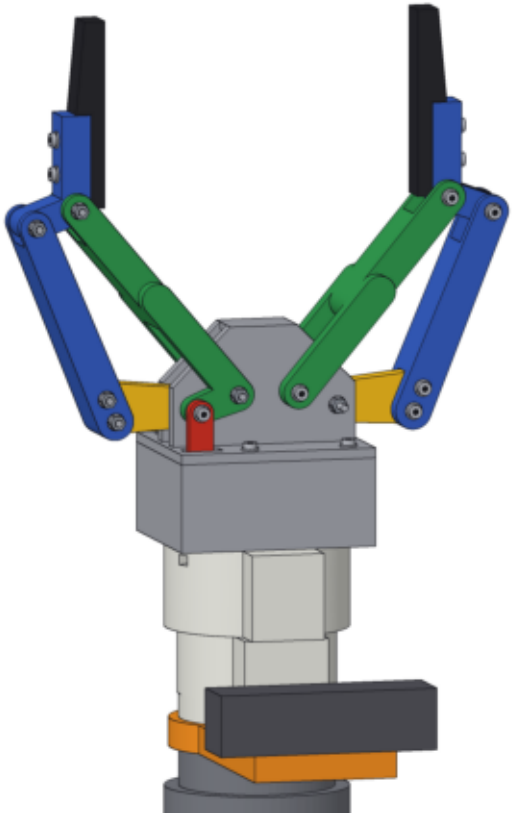
2x
M3 washer

- Move the fingers by hand until the rod hole lines up with the ear nut.

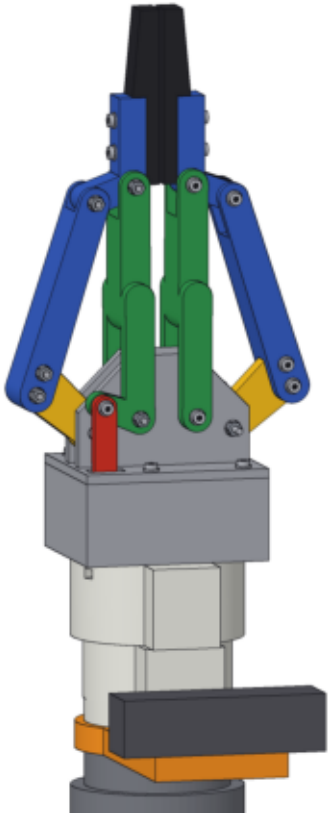
- Tighten only until the play is gone.



Done!



Open: servo 0 deg -> 130.7 mm



Closed: servo +45.4 deg -> 2 mm

First test:

- Power the servo at 6 V with a current limit.
- Software range: 0 to +45.4 deg from the assembly position (q = 0 ... 0.7 in the simulato
- Nearly linear: servo_deg = 64.8 x q.
- Mechanical stops at -2.0 and +45.7 deg.

If something rubs: check that both gears are symmetric (step 13) and that every pivot turns freely.